

Homework #14

[5.39]

$$a) T_s = \frac{1}{8 \times 10^3} = 125 \text{ MS}$$

$$T_i = \frac{T_s}{25} = 5 \text{ MS}$$

$$T_{\text{spacing}} = T_i - T_d = (5 - 1) \text{ MS} = 4 \text{ MS}$$

$$b) T_{s2} = \frac{1}{6.8 \times 10^3} = 147 \text{ MS}$$

$$T_{i2} = \frac{T_{s2}}{25} = 5.88 \text{ MS}$$

$$T_{\text{spacing}2} = T_{i2} - T_d = (5.88 - 1) \text{ MS} = 4.88 \text{ MS}$$

$$[5.48] x_1(t) = m_1(t) [\cos(\omega_a t) + \cos(\omega_b t)]$$

$$x_2(t) = m_2(t) [\cos(\omega_a t + \alpha_1) + \cos(\omega_b t + \beta_1)]$$

$$x_3(t) = m_3(t) [\cos(\omega_a t + \alpha_2) + \cos(\omega_b t + \beta_2)]$$

$$x_4(t) = m_4(t) [\cos(\omega_a t + \alpha_3) + \cos(\omega_b t + \beta_3)]$$

$$s(t) = x_1(t) + \dots + x_4(t) = \sum m_i(t) [\cos(\omega_a t + \alpha_{i-1}) + \cos(\omega_b t + \beta_{i-1})]$$

$$\alpha_0 = \beta_0 = 0 \Rightarrow r_j(t) = s(t) [\cos(\omega_a t + \alpha_{j-1}) + \cos(\omega_b t + \beta_{j-1})]$$

$$j=1, 2, 3, 4 \Rightarrow r_j(t) = \sum_{i=1}^4 m_i(t) [\cos(\omega_a t + \alpha_{i-1}) + \cos(\omega_b t + \beta_{i-1})] [\cos(\omega_a t + \alpha_{j-1}) + \cos(\omega_b t + \beta_{j-1})]$$

$$r_j(t) = \sum_i m_i(t) [\cos(\omega_a t + \alpha_{i-1}) \cos(\omega_a t + \alpha_{j-1}) + \cos(\omega_a t + \alpha_{j-1}) \cos(\omega_b t + \beta_{j-1}) + \cos(\omega_b t + \beta_{j-1}) \cos(\omega_a t + \alpha_{i-1}) + \cos(\omega_b t + \beta_{j-1}) \cos(\omega_b t + \beta_{i-1})]$$

$$\Rightarrow r_j(t) = \frac{1}{2} \sum m_i(t) [\cos(2\alpha) + \cos(\alpha_i - \alpha_{j-1}) + \cos(\alpha_i - \beta_{j-1}) + \cos(\beta_i - \beta_{j-1})]$$



$$\left[\cos(\alpha + \beta) + \cos(\alpha - \beta) + \cos(2\beta) + \cos(\beta - \beta) \right]$$

$$0, \quad 2\omega_a, \quad 2\omega_b, \quad \omega_a + \omega_b, \quad \omega_a - \omega_b$$

$$d_j(t) = \frac{1}{2} \sum_{i=1}^4 m_i(t) \left(\cos(\alpha_{i-1} - \alpha_{j-1}) + \cos(\beta_{i-1} - \beta_{j-1}) \right)$$

$$j=1 \quad d_1 = \frac{1}{2} \sum_{i=1}^4 m_i(t) (\cos \alpha_{i-1} + \cos \beta_{i-1})$$

$$d_1 = \frac{1}{2} \sum_{i=1}^4 m_i(t) (\cos \alpha_{i-1} + \cos \beta_{i-1})$$

$$= \frac{1}{2} \left(m_1(t) \overset{\cos}{(\alpha_0 + \beta_0)} + m_2(t) \overset{\cos}{(\alpha_1 + \beta_1)} \right.$$

$$\left. + m_3(t) \overset{\cos}{(\alpha_2 + \beta_2)} + m_4(t) \overset{\cos}{(\alpha_3 + \beta_3)} \right)$$

$$\cos \alpha_i + \cos \beta_i = \dots = 0$$

$$d_j(t) = \frac{1}{2} \sum_{i=1}^4 m_i(t) (\cos(\alpha_{i-1} - \alpha_{j-1}) + \cos(\beta_{i-1} - \beta_{j-1}))$$

$$\Rightarrow \cos(\alpha_{i-1} - \alpha_{k-1}) + \cos(\beta_{i-1} - \beta_{k-1}) = \begin{cases} 2, & i=k \\ 0, & i \neq k \end{cases}$$

